DE–8024 **13**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2014.

CLASSICAL ALGEBRA

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each questions carries equal marks.

$(5 \times 20 = 100)$

1. (a) Explain the first five terms of the series

$$\sum_{n=1}^{\infty}\left(\frac{2n^2+1}{2n^2-1}\right).$$ (10)

(b) Prove that the harmonic series $\sum \frac{1}{n^p}$ convergence if $p > 1$ and diverges if $p \leq 1$. (10)

2. (a) Test the convergence of the series $\Sigma\sqrt{\frac{n}{n+1}}\, x^n$, where x is any positive real number. (10)

(b) State and explain 'Leibnitz's test' in details. (10)

3. (a) Find the co-efficient of x^2 in the expansion of $\dfrac{1 + 2z - 3x^2}{e^x}$. (10)

(b) (i) Examine the Binomial series

$$1 + \frac{m}{1!}x + \frac{m(m-1)}{2!}x^2 + \cdots + \frac{m(m-1)\cdots(m-n+1)}{n!}x^n + \cdots.$$

(10)

(ii) Examine the logarithmic series

$$x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + (-1)^{n-1}\frac{x^n}{n} + \cdots.$$

4. (a) Sum the series

$$1 + \frac{1+3}{2!} + \frac{1+3+3^2}{3!} + \frac{1+3+3^2+3^3}{4!} + \text{ to } \infty.$$

(10)

(b) Let $\Sigma a_n x^n$ be the given power series. Let $\alpha = \limsup |a_n|^{1/n}$ and $R = \dfrac{1}{\alpha}$. (If $\alpha = 0$, $R = \infty$ and if $\alpha = \infty$, $R = 0$). Then $\Sigma a_n x^n$ converges absolutely if $|x| < R$. If $|x| > R$ the series is not convergent. (10)

5. (a) Solve the equation $6x^4 - 3x^3 + 8x^2 - x + 2 = 0$ given that it has a pair of roots whose sum is zero. (10)

(b) If α, β, γ are the roots of $x^3 + qx + r = 0$, find the value of $\dfrac{1}{\beta+\gamma} + \dfrac{1}{\gamma+\alpha} + \dfrac{1}{\alpha+\beta}$. (10)

6. (a) Solve $x^4 + 16x^3 + 83x^2 + 152x + 84 = 0$ by removing the second term. (10)

(b) Solve the equation
$6x^5 + 11x^4 - 37x^3 - 33x^2 + 11x + 6 = 0$. (10)

7. (a) If $x^2 = y^2 + z^2$, show that $x^n \geq y^n + z^n$ whenever $n \geq 2$. (10)

(b) If $a, b, c, \cdots k$ are positive numbers, then

$$\frac{a^m + b^m + c^m + \cdots + k^m}{n} > \frac{(a + b + c + \cdots + k)^m}{n}$$

except, when m lies between 0 and 1. (10)

8. (a) Diagonalise the matrix $\begin{bmatrix} 4 & -4 & 6 \\ 2 & 2 & 2 \\ 2 & 6 & -2 \end{bmatrix}$. (10)

(b) Solve the following simultaneous equations. (10)

$2x + 4y + 10z = 46$

$6x + 2y + 8z = 52$

$12x + 2y + 14z = 94$

DE–8025 **14**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION,
MAY 2014.

CALCULUS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) If $y = \sin(m \sin^{-1} x)$, prove that

$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2 - n^2)y_n = 0$. Hence show that $y_{n+2}(0) = (n^2 - m^2)y_n(0)$

(b) If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, show that

(i) $\dfrac{\partial u}{\partial x} + \dfrac{\partial u}{\partial y} + \dfrac{\partial u}{\partial z} = \dfrac{3}{x+y+z}$

(ii) $\left(\dfrac{\partial}{\partial x} + \dfrac{\partial}{\partial y} + \dfrac{\partial}{\partial z}\right)^2 (u) = \dfrac{-9}{(x+y+z)^2}$.

2. (a) Prove that the $p-r$ equation of the cardiod $r = a(1-\cos\theta)$ is $p^2 = \dfrac{r^3}{2a}$.

(b) Find the radius of curvature of the curve $x^3 + 2x^2y + 3xy^2 - 4y^3 + 5x^2 - 6xy + 7y^2 - 8y = 0$ at (0,0)

3. (a) (i) Evaluate $\int \dfrac{dx}{x^2 - a^2}$

(ii) Evaluate $\int \dfrac{dx}{a^2 - x^2}$

(b) Evaluate $\int x^3 e^{2x} dx$.

4. (a) Establish a reduction formula for $I_n = \int \sin^n x dx$ where $n \in N$ and hence find $\int_0^{\frac{\pi}{2}} \sin^n x\, dx$.

(b) If $f(m,n) = \int_0^{\frac{\pi}{2}} \cos^m x \sin nx\, dx$, prove that $f(m,n) = \frac{1}{m+n} + \frac{m}{m+n} f(m-1, n-1)$. Hence deduce that $f(m,n) = \frac{1}{2^{m+1}} \left[\frac{2}{1} + \frac{2^2}{2} + \frac{2^3}{3} + \ldots + \frac{2^m}{m} \right]$.

5. (a) Solve $\frac{dy}{dx} - y \cot x = 2x \sin x$

(b) Solve $y = px + p - p^2$.

6. (a) Solve $(D^2 - 4)y = e^{2x} + e^{-4x}$.

(b) Solve $(D^3 - 3D^2 + 3D - 1)y = x^2 e^x$.

7. (a) Find $L\left(\frac{1 - \cos x}{x}\right)$.

(b) Find $L^{-1}\left[\frac{1}{S(S+1)(S+2)}\right]$.

8. (a) Solve $P \cot x + q \cot y = \cot z$.

(b) Find the complete integral of $px + qy = pq$.

DE–8026 **15**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2014.

ANALYTICAL GEOMETRY AND VECTOR CALCULUS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) Show that two of the straight lines $ax^3 + bx^2y + cxy^2 + dy^3 = 0$ will be perpendiculars to each other if $a^2 + d^2 + bd + ac = 0$. (10)

   (b) A variable circle is drawn to touch the y-axis at the origin. Find the locus of the pole of the straight line $lx + my + n = 0$ with respect to it. (10)

2. (a) Find the circles which cut orthogonally each of the following circles : (10)

$$x^2 + y^2 + 2x + 4y + 1 = 0$$
$$x^2 + y^2 - 4x + 3 = 0$$
$$x^2 + y^2 + 6y + 5 = 0$$

   (b) Prove that the tangents at the extremities of any focal chord of a conic interest on the corresponding directrix. (10)

3. (a) If the direction cosines of two lines satisfy $l+m+n=0$, and $2lm+2mn-nl=0$, then find the direction cosines. (10)

(b) Prove : Length of the perpendicular from a point $A(x_1, y_1, z_1)$ to the plane $ax+by+cz+k=0$. Is $\pm\left[\frac{ax_1+bx_1+cz_1+d}{\sqrt{(a^2+b^2+c^2)}}\right]$. (10)

4. (a) Find the equation of the plane passing through the points $(3,-8,2)$ and $(-2,-1,1)$ and perpendicular to the plane $x+y+z+1=0$. (10)

(b) Find the image of the point $(2,3,4)$ under the reflection in the plane $x-2y+5z=6$. (10)

5. (a) Prove that $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-2}{3}=\frac{y-3}{4}=\frac{z-4}{5}$ are coplanar and find the equation of the plane containing them. (10)

(b) Prove : The lengths of two opposite edges of a tetrahedron are a, b; their shortest distance is equal to d and the angle between them is θ. Prove that its volume is $\frac{abd\sin\theta}{6}$. (10)

6. (a) Prove that a plane section of a sphere is a circle. (10)

(b) Find the equation of tangent plane at a point x_1, y_1, z_1 to the sphere

$x^2+y^2+z^2+2ux+2vy+2wz+d+0$. (10)

7. (a) Show that $\overline{F} = (6xy + z^3)\ \bar{i} + (3x^2 - z)\ \bar{j} + (3xz^2 - y)\overline{k}$ is irrotational and find ϕ such that $\overline{F} = \nabla\ \phi$. (10)

(b) Prove that $div(\overline{A} \times \overline{B}) = \overline{B}.\ curl\ \overline{A} - \overline{A}.\ curl\ \overline{B}$. (10)

8. (a) Use Gauss divergence theorem find $\iint_s \overline{F}.\ n\, ds$ where $\overline{F} = x^3\ \bar{i} + y^3\ \bar{j} + z^3\ \overline{h}$ and S is the surface of the sphere $x^2 + y^2 + z^2 = a^2$. (10)

(b) Verify Stoke's theorem for

$f = (2x - y)i - y\,z^2 j - y^2 z\,k$ where S in upper half surface of the sphere $x^2 + y^2 + z^2 = 1$ and C is its boundary. (10)

DE–8027 **23**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2014.

MECHANICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) The resultant of two forces P,Q acting at a certain angle is X and that P,R acting at the same angle is also X. The resultant of Q,R again acting at the same angle is Y. Prove that

$$P = \left(X^2 + QR\right)^{1/2} = \frac{QR(Q+R)}{Q^2 + R^2 - y^2}$$

Prove also that if $P + Q + R = 0$, $Y = X$. (10)

(b) ABC is a given triangle, Forces P,Q,R acting along the lines OA, OB, OC are in equilibrium. Prove that P : Q : R = $a^2\left(b^2 + c^2 - a^2\right)$: $b^2\left(c^2 + a^2 - b^2\right)$: $c^2\left(a^2 + b^2 - c^2\right)$ if 0 is the circumcentre of the triangle. (10)

2. (a) State and prove Lami's theorem. (10)

(b) OA, OB, OC are the lines of action of two forces P and Q and their resultant R respectively. Any transversal meets the lines in L,M and N respectively. Prove that $\frac{P}{OL} + \frac{Q}{OM} = \frac{R}{ON}$. (10)

3. (a) Find the resultant of two unlilke and unequal parallel forces acting on a rigid body. (10)

(b) If D is any point on the base BC of triangle ABC such that $\frac{BD}{DC} = \frac{m}{n}$ and $\angle ADC = \theta$, $\angle BAD = \alpha$ and $\angle DAC = \beta$ then prove the following

(i) $(m+n)\cot\theta = m\cot\alpha - n\cot\beta$

(ii) $(m+n)\cot\theta = n\cot B - m\cot C$. (10)

4. (a) A particle of weight 30 kgs resting on a rough horizontal plane is just on the point of motion when acted on by horizontal forces of 6 kgwt and 8 kgwt at right angles to each other. Find the coefficient of friction between the particle and the plane and the direction in which the friction acts. (10)

(b) A beam of weight W hinged at one end is supported at the other end by a string so that the beam and the string are in a vertical plane and make the same angle θ with the horizon. Show that the reaction at the hinge is $\frac{W}{4}\sqrt{8+\operatorname{cosec}^2\theta}$. (10)

5. (a) Show that the greatest height which a particle with initial velocity v can reach on a vertical wall at a distance 'a' from the point of projection is $\frac{v^2}{2g} - \frac{ga^2}{2v^2}$. (10)

(b) The range of a rifle bullet is 1000 m, when α is the angle of projection. show that if the bullet is fired with the same elevation from a car travelling 36 km/h towards the target, the range will be increased by $\frac{1000\sqrt{\tan\alpha}}{7}$ m. (10)

6. (a) A shot of mass m penetrates a thickness t of a fixed plate of mass M. If M were free to move and the resistance supposed to be uniform. Show that the thickness penetrated is $\frac{Mt}{M+m}$. (10)

(b) A particle is projected from a point on an inclined plane and at the rth impact it strikes the plane perpendicularly and at the nth impact is at the point of projection, show that $e^n - 2e^r + 1 = 0$. (10)

7. (a) A particle is moving with S.H.M and while making an oscillation from one extreme position to the other, its distances from the centre of oscillation at 3 consecutive seconds are x_1, x_2, x_3. Prove that the period of oscillation is $\frac{2\pi}{Cos^{-1}\left(\frac{x_1 + x_3}{2x_2}\right)}$. (10)

(b) If the displacement of a moving point at any time be given by an equation of the form $x = a\cos wt + b\sin wt$, show that the motion is a simple harmonic motion. If $a = 3$, $b = 4$, $w = 2$ determine the period, amplitude, maximum velocity and maximum acceleration of the motion. (10)

8. (a) Discuss the composition of two simple harmonic motions of the same period and in the same straight lines. (10)

(b) Show that the path of a point p which possesses two constant velocities u and v, the first of which is in a fixed direction and the second of which is perpendicular to the radius OP drawn from a fixed point O, is a conic whose focus is O and whose eccentricity is $\frac{u}{v}$. (10)

DE–8028 **24**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2014.

ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Let $x, y \in R^2$. Then $x = (x_1, x_2)$ and $y = (y_1, y_2)$ where $x_1, x_2, y_1, y_2 \in R$. We define $d(x, y) = |x_1 - y_1| + |x_2 - y_2|$ then prove that d is a metric on R^2.

   (b) Prove that in any metric space (M, d), each open ball is an open set.

   (c) Let (M, d) be a metric space. Let $x \in M$. Show that $\{x\}^c$ is open. (8 + 8 + 4 = 20)

2. (a) Let M be a metric space and M_1 a subspace of M. Let $A_1 \subseteq M_1$. Then prove that A_1 is open in M_1 if and only if there exists an open set A in M such that $A_1 = A \cap M_1$.

   (b) Let (M, d) be a metric space. Let $A \subseteq M$. Then prove that x is a limit point of A iff each open ball with centre x contains an infinite number of points of A. (10 + 10 = 20)

3. (a) Prove that C with usual metric is complete.

(b) State and prove Cantor's Intersection theorem. (10 + 10 = 20)

4. (a) Prove that a subspace of R is connected if and only if it is an interval.

(b) Prove that the closure of a totally bounded set is totally bounded. (12 + 8 = 20)

5. (a) Prove that any compact subset A of a metric space M is bounded.

(b) Prove that a metric space M is compact if and only if any family of closed sets with finite intersection property has non empty intersection. (8 + 12 = 20)

6. (a) Let M be a metric space let $A \subseteq M$. Prove that A is compact iff given a family of open sets $\{G_\alpha\}$ in M such that $UG_\alpha \supseteq A$ there exists a subfamily $G_{\alpha_1}, G_{\alpha_2}, \cdots G_{\alpha_n}$ such that $\bigcup_{i=1}^{n} G_{\alpha_i} \supseteq A$.

(b) Prove that any continuous real valued function f defined on a compact metric space is bounded and attains its bounds. (12 + 8 = 20)

7. (a) Consider $f_n : R \to R$ defined by $f_n(x) = \dfrac{nx}{1+n^2x^2}$. Determine whether the convergence is uniform or not.

(b) Prove that the uniform limit of a sequence of continuous functions is continuous.

(c) Let $T : M \to M$ be a contraction mapping. Then prove that T is continuous on M. (8 + 8 + 4 = 20)

8. (a) State and prove Piccard's theorem.

(b) Let $f_n : [0,1] \to R$ be defined by $f_n(x) = \frac{x}{n}$. Prove that the sequence (f_n) converge pointwise to $f : [0,1] \to R$ defined by $f(x) = 0$. (12 + 8 = 20)

DE–8029 **25**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2014.

PROBABILITY AND STATISTICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) Prove : If A and B are any two events of a random experiment with sample space S then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. (10)

   (b) The probability that India wins a cricket test match against West Indies is known to be 2/5. If India and West Indies play three test matches what is the probability that :

   (i) India will loose all the three test matches.

   (ii) India will win atleast one test match.

   (iii) India will win all tests.

   (iv) India will win atmost one match. (10)

2. (a) Obtain the

   (i) Mean

   (ii) Median and

   (iii) Mode for the following distribution :

$$f(x) = \begin{cases} 6(x - x^2) & \text{if } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$ (10)

(b) Find the correlation coefficient from the following data : (10)

X :	10	12	18	24	23	27
Y :	13	18	12	25	30	10

3. (a) If X, Y and Z are uncorrelated random variables each having same standard deviation obtain the coefficient of correlation between the r.v's $X + Y$ and $Y + Z$. (10)

(b) Let x_1, x_2, ..., x_n be the ranks of n individuals according to a character A and y_1, y_2, ..., y_n the ranks of the same individuals according to another character B. It is given that $x_i + y_i = 1 + n$ for $i = 1, 2, ..., n$. Show that the value of the rank correlation ρ between the character A and B is –1. (10)

4. (a) The random variables X and Y are connected by the equation $ax + by + c = 0$. Show that $\gamma_{XY} = -1$ or 1 according as a and b are of the same sign or of opposite sign. (10)

(b) Find the correlation coefficient between X and Y from the following table : (10)

y \ x	5	10	15	20
4	2	4	5	4
6	5	3	6	2
8	3	8	2	3

5. (a) Find β and γ coefficients for the Binomial distribution and discuss the results with special reference to skewness and Kurtosis. (10)

(b) Assuming that one in 80 births is a case of twins calculate the probability of 2 or more births of twins on a day when 30 births occur using

(i) Binomial distribution and

(ii) Poisson distribution. (10)

6. (a) A random variable X has the probability function $p(x) = \frac{1}{2}$, $x = 1, 2, 3, \ldots$ find its moment generating function and variance. (10)

(b) In a normal distribution 31% of the items are under 45 and 8% are over 64. Find the mean (μ) and standard deviation (σ). (10)

7. (a) The theory predicts that the proportion of an object available in four groups A, B, C, D should be 9 : 3 : 3 : 1. In an experiment among 1600 items of this object the numbers in the numbers in the four groups were 882, 33, 287 and 118. Use χ^2 – test to verify whether the experimental results supports the theory. (10)

(b) The table gives the biological values of protein from 6 cow's milk and 6 buffalo's milk. Examine whether the differences are significant. (10)

Cow's milk	Buffalo's milk
1.8	20
2.0	1.8
1.9	1.8
1.6	2.0
1.8	2.1
1.5	1.9

8. (a) The following is the statistics showing the life times in hours of four batches of electric bulbs sold in different shops. Perform an analysis of variance and state your conclusion. (10)

Batches	S_1	S_2	S_3	S_4	S_5	S_6	S_7	S_8
A	1600	1610	1650	1680	1700	1720	1800	–
B	1580	1640	1640	1700	1750	–	–	–
C	1460	1550	1600	1620	1640	1660	1740	1820
D	1510	1520	1530	1570	1600	1680	–	–

(b) Calculate :

(i) Laspeyre's

(ii) Paasches'

(iii) Fishers' index numbers for the following data.

Hence or otherwise find Edgeworth and Bowley's index numbers. (10)

Commodities	Base Year 1990		Current year 1992	
	Price	Quantity	Price	Quantity
A	2	10	3	12
B	5	16	6.5	11
C	3.5	18	4	16
D	7	21	9	25
E	3	11	3.5	20

DE–8030 **31**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2014.

ALGEBRA

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) If A, B and C are any three finite sets prove that $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|$. (10)

   (b) If A, B and C are any three sets prove that

   (i) $(A \cap B) \times C = (A \times C) \cap (B \times C)$

   (ii) $A \times (B - C) = (A \times B) - (A \times C)$. (10)

2. (a) If ρ and σ are equivalence relations defined on a set S, prove that $\rho \cap \sigma$ is also an equivalence relation. (10)

   (b) Let $f : X \to Y$ be a function. If $A \subseteq X$ and $B \subseteq Y$ show that :

   (i) $A \subseteq f^{-1}[f(A)]$ and

   (ii) $f[f^{-1}(B)] \subseteq B$.

   (iii) Give an example to show that equality need not hold in (i) and (ii).

   (iv) In each case state when will the equality hold? (10)

3. (a) Prove : Let G be a group and $a, b \in G$, then

(i) order of a = order of a^{-1}.

(ii) order of a = order of $b^{-1}ab$.

(iii) order of ab = order of ba. (10)

(b) Let H and K be two subgroups of G of finite index in G. Prove that $H \cap K$ is a subgroup of finite index in G. (10)

4. (a) State and prove 'Lagrange's theorem' on group. (10)

(b) State and prove 'Fundamental theorem' of homomorphism. (10)

5. (a) Let Z be the ring of integers and R any ring. Define $Z \times R = \{(m,x) / m \in Z \text{ and } x \in R\}$, $\oplus$ and $\odot$ on $Z \times R$ as follows $(m.x) \oplus (n.y) = (m+n, x+y)$; $(m.x) \odot (n.y) = (mn.my + nx + xy)$ where nx and my denotes respectively the concerned multiples of the elements x and y in R. Prove that $Z \times R$ is a ring under $\oplus$ and $\odot$. Also prove that $Z \times R$ is commutative iff R is commutative. (10)

(b) Prove : The field of quotients F of an integral domain D is the smallest field containing D. (ie) If F' is any other field containing D then F' contains a subfield isomorphic to F. (10)

6. (a) Prove that the ring of Gaussian integers $R = \{a + bi / a, b \in Z\}$ is an Eucledian domain, where $d(a+bi) = a^2 + b^2$. (10)

(b) State and prove 'Division algorithm'. (10)

7. (a) Let V be a vector space over a field F. Let S, $T \subseteq V$, prove that

(i) $S \subseteq T \Rightarrow L(S) \subseteq L(T)$

(ii) $L(S \cup T) = L(S) + L(T)$

(iii) $L(S) = S$ iff S is a subspace of V. (10)

(b) Prove : Let V be a vector space over a field F. Let $S = \{V_1, V_2, ..., V_n\}$ spans V, and $S = \{W_1, W_2, ..., W_n\}$ be a linearly independent set of vectors in V, then $m \leq n$. (10)

8. (a) Prove that every finite dimensional inner product space has orthonormal basis. (10)

(b) Prove : Let V be a finite dimensional inner product space, and W be a subspace of V, then V is the direct sum of W and $W^{\perp}$ (ie) $V = W \oplus W^{\perp}$. (10)

DE–8031 **32**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2014

OPERATIONS RESEARCH

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) Solve the graphically the LPP;
minimize $Z = 2x + 3y$,
Subject to : $x + y \leq 30$,
$x - y \geq 0$,
$y \geq 0$,
$0 \leq x \leq 20, 0 \leq y \leq 12$ (10)

(b) Explain Simplex Method to find an optimal basic feasible solution to LPP. (10)

2. (a) A farmer has 100 acres of land. He cultivates tomato, brinjal and green chilly in this land. He can sell all vegetable in the market and earn a profit of Rs. 2per kg on tomatoes and brinjal and Rs .3 per kg on green chillies . The average yield per acre in 2000 kgs of tomatoes, 3000 kgs of brinjals and 500 kgs of green chilly. Fertilizers are available at Rs. 5 per kg and the amount required per acre is 100 kgs each for tomatoes, brinjals and 50 kgs for green chillies. Labour required for sowing, cultivating and harvesting per acre is 5 man days for tomatoes, 4 man days for brinjal and 6 man days of labour are available at Rs. 60 per man-day. Formulate this problem as an LPP to maximize the farmer's total profit. (10)

(b) Use Simplex method to solve the following LPP :

Maximize $z = 2x_1 + 3x_2$,

Subject to the constrains :

$-x_1 + 2x_2 \le 4$

$x_1 + x_2 \le 6$

$x_1 + 3x_2 \le 9$, x_1, x_2 unrestricted. (10)

3. (a) Use the penalty method to solve the following LPP.

Maximize $z = 2x_1 + 3x_2$

Subject to the constrains :

$x_1 + 2x_2 \le 4$ and

$x_1 + x_2 = 3$, $x_1 \ge 0$ and $x_2 \ge 0$ (10)

(b) Solve the following LPP using Two-phase Simplex method :

Minimize $Z = 60x_1 + 80x_2$,

Subject to : $20x_1 + 30x_2 \ge 900$

$40x_1 + 30x_2 \ge 1200$, $x_1, x_2 \ge 0$ (10)

4. (a) Explain Two-phase Simplex method algorithm. (10)

(b) Find the integer solution to the L.P.P. :

Maximize $z = 2x_1 + 2x_2$

Subject to the constrains :

$5x_1 + 3x_2 \le 8$,

$x_1 + 2x_2 \le 4$, $x_1, x_2 \ge 0$ and are integers. (10)

5. (a) Using Row Minima method find a basic feasible solution to the following transportation problem : (10)

	w_1	w_2	w_3	ai
F_1	8	10	12	900
F_2	12	13	12	1000
F_3	14	10	11	1200
bj	1200	1000	900	3100

(b) A company produces a small component for an industrial product and distributes it to five wholesale at a fixed delivered price of Rs. 2.50 per unit. Sales forecast indicate that monthly deliveries will be 3000, 3000,10000, 5000 and 4000 units to wholesalers 1,2,3,4 and 5 respectively. The monthly production capacities are Re. 1.00, Re 0.90 and Re. 0.80 at plants 1, 2 and 3 respectively. The transportation costs of shipping a unit from a plant to a wholesaler are given below :

		Wholesaler				
		1	2	3	4	5
	1	.05	.07	.10	.15	.15
Plant	2	.08	.06	.09	.12	.14
	3	.10	.09	.08	.10	.15

Find how many components each plant supplies to each wholesaler in order to maximize its profit. (10)

6. A departmental head has four subordinates, and four tasks to be performed. The subordinates differ in efficiency. And the tasks differ in their intrinsic difficulty. His estimate, of the time each man would take to perform each tasks, is given in the matrix below :

Tasks	Men			
	E	F	G	H
A	18	26	17	11
B	13	28	14	26
C	38	19	18	15
D	19	26	24	10

How the tasks should be allocated one to a man, so as to minimize the total man-hours? (20)

7. (a) Use graphical method in solving the following game (10)

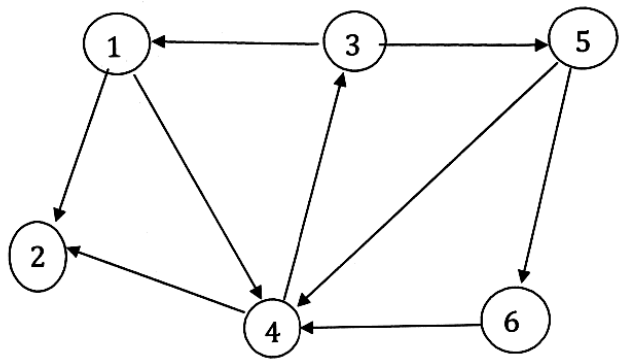
$$\begin{array}{cc} & \text{Player A} \\ \text{Player B} & \begin{pmatrix} 2 & 2 & 3 & -2 \\ 4 & 3 & 2 & 6 \end{pmatrix} \end{array}$$

(b) (i) Define zero-sum game. (5)

(ii) Define payoff matrix. (5)

8. (a) Discuss about PERT algorithm. (10)

(b) For the following directed network



(i) Determine a directed path from node 1 to node 6. Identify three undirected paths connecting node 1 and node 6.

(ii) Find three directed cycles.

(iii) Grow a tree one link at a time until a spanning tree has been obtained.

(iv) Repeat the process of (c) to obtain another spanning tree. (10)

DE–8032 **33**

DISTANCE EDUCATION

B.Sc. (Mathematics). DEGREE EXAMINATION, MAY 2014

NUMERICAL METHODS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Find a real roots of the equation $x^3 - x - 11 = 0$ by using bisection method. (10)

   (b) (i) Explain the method of false position is solving an algebraic equations. (5)

   (ii) Evaluate $\sqrt{12}$ to four places of decimals by Newton – Rapshon method. (5)

2. (a) Find the inverse of the matrix $A = \begin{pmatrix} 2 & 1 & 1 \\ 3 & 2 & 3 \\ 1 & 4 & 9 \end{pmatrix}$ using Gaussian method. (10)

   (b) Solve the system of equations by Gauss elimination method : (10)

   $3x + y + 2z = 3$
   $2x - 3y - z = -3$
   $x + 2y + z = 4$

3. (a) Using Newton's forward interpolation find $f(0.2)$ from the following data : (10)

x :	0	1	2	3	4	5	6
$Y = f(x)$:	176	185	194	203	212	220	229

(b) Define Langrange's intepolation formula. (10)

4. (a) Using Bessel's formula find $Y(25)$ from the given data : (10)

X :	20	24	28	32
Y :	2854	3162	3544	3992

(b) Using Hermite's Interpolation to find sin (1.05) from the following data : (10)

X :	1.0	1.1
$Y = \sin x$:	0.84147	0.89121
$Y' = \cos x$:	0.5403	0.45360

5. (a) Derive Trapezoidal rule for integration. (10)

(b) Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using Simpson's percentage rule. (10)

6. (a) Derive Weddle's rule for integral. (10)

(b) Evaluate $\int_0^1 \frac{dx}{1+x}$, using Weddle's rule. (10)

7. Using Runge–Kutta method to find $y(0.1)$, $y(0.2)$ and $y(0.3)$ from the differential equation $\frac{dy}{dx} = 1 + xy$, $y(0) = 2$. (20)

8. Find an approximate solution of the initial value problem $Y' = 1 + y^2, y(0) = 0$ by Picards methods and compare it with the exact solution. (20)

DE–8033 **34**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2014.

COMPLEX ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each questions carries equal marks.

$(5 \times 20 = 100)$

1. (a) Prove that $f(z) = z$ lm z is differentiable only at $z = 0$ and find $f'(0)$. (10)

   (b) Prove that the functions $f(z)$ and $\overline{f(z)}$ are simultaneously analytic. (10)

2. (a) Show that

   $u(x, y) = \sin x \cosh y + 2\cos x \sinh y + x^2 - y^2 + 4xy$ is harmonic. Find an analytic function $f(z)$ in terms of z with the given u for its real part. (10)

   (b) Prove :

   If u and v are harmonic functions satisfying the Cauchy-Riemann equations in a region D then $f = u + iv$ is analytic in D. (10)

3. (a) Show that the region in the z - plane given by $x > 0$ and $0 < y < 2$ is mapped to the region in the w - plane given by $-1 < u < 1$ and $v > 0$ under the transformation $w = iz + 1$. (10)

   (b) Prove that any bilinear transformation can be expressed as a product of translation, rotation, magnification or contraction and inversion. (10)

4. (a) Find the bilinear transformation which maps the points $z = -1, 1, \infty$ respectively on $w = -i, -1, i$. (10)

(b) Prove let C be a circle or a straight line and z_1, z_2 be inverse points or reflection points with respect to C. Let w_1, w_2 and C_1 be the image of z_1, z_2 and C under bilinear transformation. Then w_1 and w_2 are inverse points or reflection points will respect to C_1. (10)

5. (a) Prove : Let $f(z)$ be a function which is analytic inside and on a simple. closed curve C and z_0 be any point in the interior of C, then $f(z_0) = \frac{1}{2\pi i}\int_c \frac{f(z)}{z - z_0} dz$. (10)

(b) Evaluate $\int_c \frac{z}{z^2 - 1} dz$ where C is the positively oriented circle $|z| = 2$. (10)

6. (a) State and prove Liouvilie's theorem. (10)

(b) Evaluate $\int_c \frac{z^3}{(2z + i)} dz$, where C is the unit circle $|z| = 2$. (10)

7. (a) For the function $f(z)\frac{2z^3 + 1}{z(z + i)}$, find

(i) a Taylor's series valid in a neigh-bourhood of $z = i$ and

(ii) a Laurent's series valid within an annulus of which centre is the origin. (10)

(b) Calculate the residue of $\frac{z + 1}{(z^2 - 2z)}$ at its poles. (10)

8. (a) Show that the function $2+z^2-e^{iz}$ has Precisely one zero in the open upper half plane. (10)

(b) Prove that $\int_c \frac{e^{2z}}{(z+1)^3}dz = \frac{4\pi i}{e^2}$, where C is $|Z| = \frac{3}{2}$. (10)

DE–8034 **35**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2014.

DISCRETE MATHEMATICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each questions carries equal marks.

$(5 \times 20 = 100)$

1. (a) Explain: Negation, Conjunction, Disjunction, Conditional statement and Biconditional statement. (10)

   (b) Construct a truth table for the following compound propositions:

   (i) $(p \vee q) \leftrightarrow (p \rightarrow \neg r)$

   (ii) $p \rightarrow (q \leftrightarrow (r \rightarrow s))$. (10)

2. (a) Indicate which ones are tautology or contradiction

   (i) $(q \vee r) \rightarrow (p \wedge \neg r)$

   (ii) $(P \rightarrow (Q \rightarrow R)) \Rightarrow ((P \rightarrow Q) \rightarrow (P \rightarrow R))$. (10)

   (b) Obtain the principal disjunctive normal form for

   $P \rightarrow ((P \rightarrow Q) \wedge \neg(\neg Q \vee \neg P))$. (10)

3. (a) Show that:

   $\{(p \rightarrow q) \wedge (r \rightarrow s), (q \rightarrow t) \wedge (s \rightarrow u), \neg(t \wedge u), p \rightarrow r\}$

   $q \Rightarrow \neg p$.

   (10)

(b) Determine the validity of the following argument:

My father praises me only if I can be proud of myself. Either I do well in sports or cannot be proud of myself. If I study hard, then I cannot do well in sports. Therefore, if father praises me, then I do not study well. (10)

4. (a) Show that:

$(\forall x)(F(x) \wedge S(x)) \rightarrow (\forall y)(M(y) \rightarrow W(y))$. (10)

(b) Show that $(\forall x) M(x)$ follows logically from the premises $(\forall x)(H(x) \rightarrow M(x))$ and $((\forall x) H(x))$. (10)

5. (a) Prove in any graph, the number of vertices of odd degree is always even. (10)

(b) Define the adjacency matrix and incidence matrix of a graph with an example. (10)

6. (a) (i) Prove: A connected graph with n vertices and $n-1$ edges is a tree. (5)

(ii) Define binary tree with an example. (5)

(b) Prove Let T be a full binary tree with n vertices, then the number of leaves in T is $(n+1)/2$. (10)

7. (a) Prove: Every cycle has an even number of edges in common with any cut-set. (10)

(b) Prove that if $n(\neq 1)$ is a positive odd integer, then the complete graph K_n contains $\frac{(n-1)}{2!}$ edge — disjoint Hamilton cycles. (10)

8. (a) Explain Ford and Fulkerson algorithm. (10)

(b) State Euler's formula for a graph $G = (V, E)$. (10)